

Bocconi

UNDERSTANDING HOW THE PROFIT AND LOSS ACCOUNT WORKS 30060 - MANAGEMENT

Walsh Book

Department of Management & Technology



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AGENDA

1. PROFIT AND LOSS (P/L) ACCOUNT
2. FINANCIAL MEASURES
3. P/L AND STAKEHOLDERS
4. RATIOS FOR STOCK MARKET INVESTORS



1.0

PROFIT AND LOSS ACCOUNT

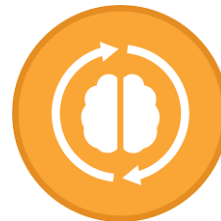


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What is the purpose of the P&L?



UNDERSTAND WHERE IMPACTS ARE



DEVELOP A KNOWLEDGEABLE DECISION-
MAKING PROCESS



COMPARISON



TRANSPARENCY WITH
STAKEHOLDERS



STRENGTHEN THE BUSINESS
AND MANAGE RISKS FOR THE
FUTURE



We define **PROFIT AND LOSS** as

Gains or losses from both normal and abnormal company operations over a period of time



Profit and loss account

- Measures total revenues minus total costs (=operating profit) over the period between the opening and closing balance sheets of an accounting period



B/S 31 Dec. 2019



P&L account



B/S 31 Dec. 2020

2.0

FINANCIAL MEASURES



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List of financial measures

	REVENUES
	OPERATING COSTS
-	(Labor, Materials, Overheads, Administration, Sales and Marketing)
=	EBITDA (Earnings Before Interests, Taxes, Depreciations, Amortizations)
-	DEPRECIATIONS
-	AMORTIZATIONS
=	EBIT (Earnings Before Interests, Taxes)
-	INTERESTS
=	EBT (Earnings Before Taxes)
-	TAXES
=	NET INCOME (or Earnings After Taxes)
-	DIVIDENDS
=	RETAINED EARNINGS



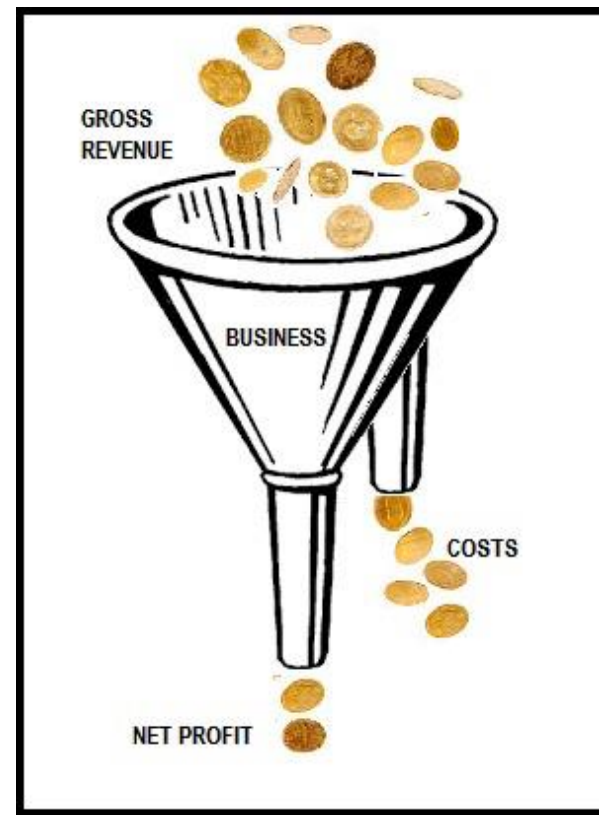
List of financial measures

	REVENUES
	OPERATING COSTS
-	(Labor, Materials, Overheads, Administration, Sales and Marketing)
=	EBITDA (Earnings Before Interests, Taxes, Depreciations, Amortizations)

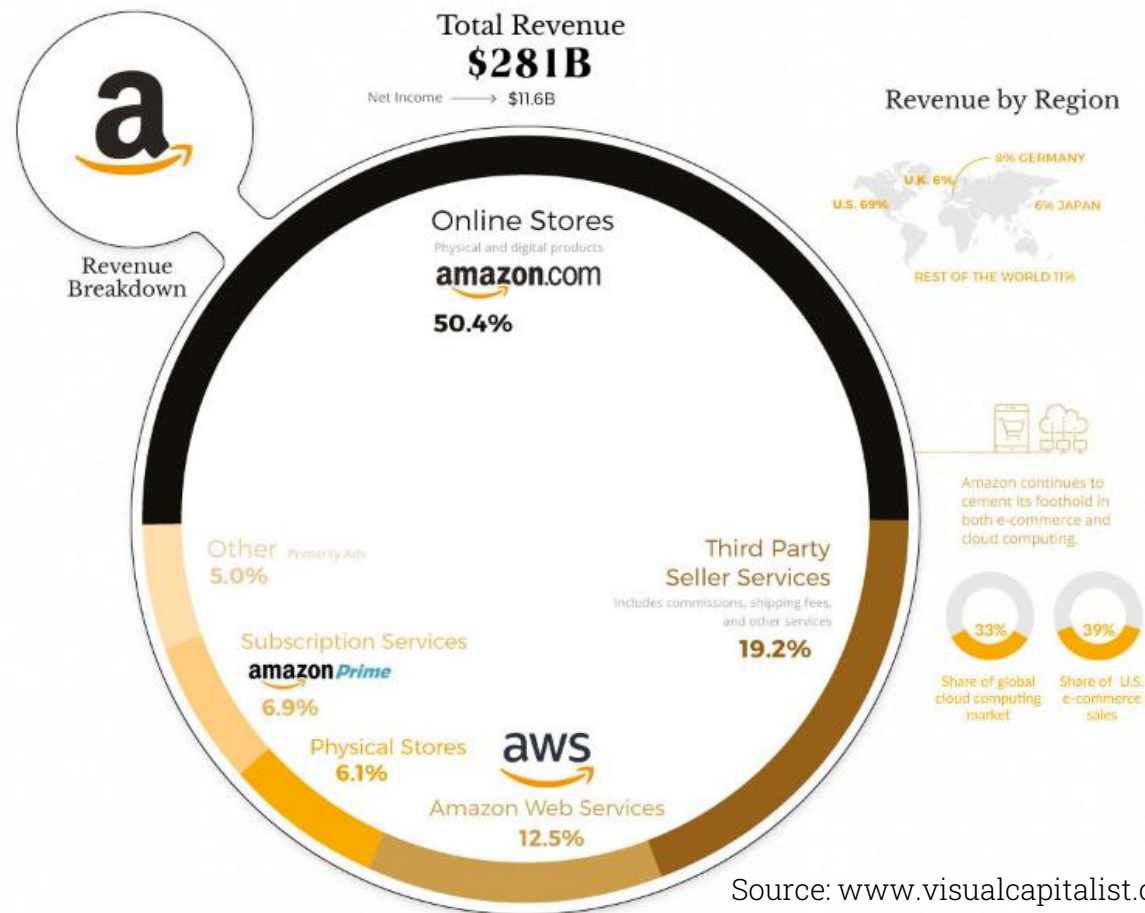


Revenues

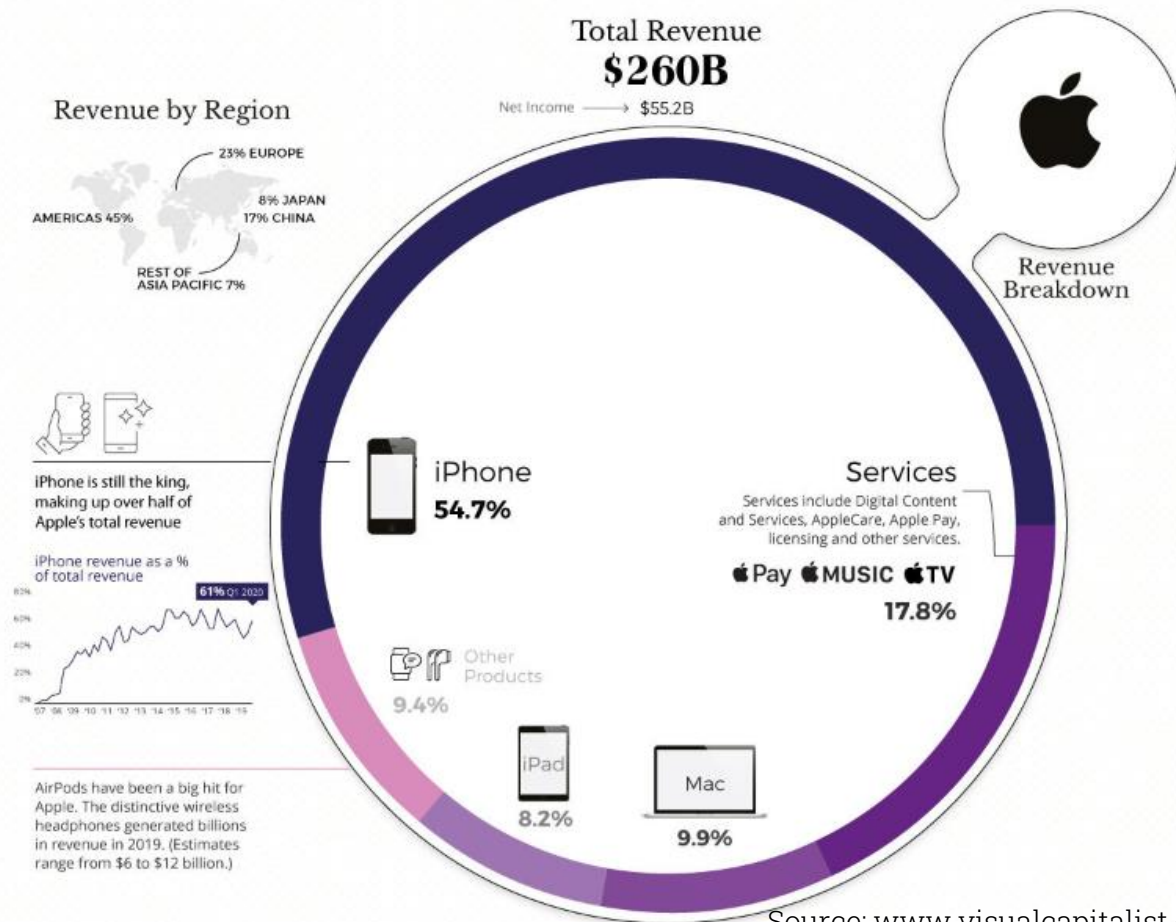
- They mainly refer to the value of a company's sales of goods and services to its customers
- Even though a company's "bottom line" (its net income) gets most of the attention from investors, the "top line" is where the revenue or income process begins
- Companies typically can grow no faster than their revenues



Amazon revenue breakdown for 2020

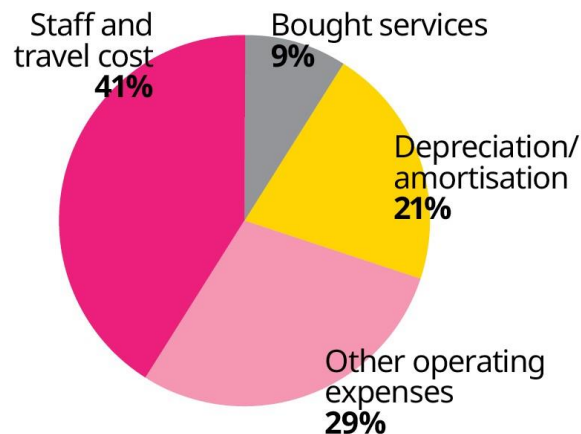


Apple revenue breakdown for 2020



Operating costs

- For a manufacturer, are the expenses incurred for raw materials, labor and manufacturing overhead used in the production of its goods
- For retailers, are essentially the purchase cost of merchandise used for resale
- For service-related firms, are the cost of services rendered or cost of revenues

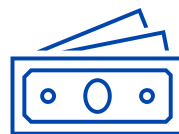


Ikea operating costs (source: www.ikea.com)

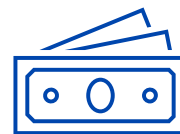


Overhead costs

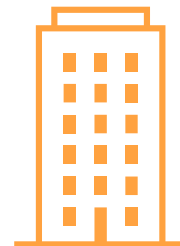
- All ongoing expenses that are not directly related to the product or service being produced (i.e. direct material/direct labor)
- Overhead are paid regardless of whether a company is performing well (i.e. rent, insurance, utilities, office supplies, salaries to human resources department and to accounting department)



RENT



Direct cost



Indirect cost



EBITDA

- Earnings Before Interests, Taxes, Depreciations, Amortizations
- It represents the cash that is thrown off from the normal trading activities of the business
- EBITDA is a number that is largely free from distortions that can arise from accounting policies.

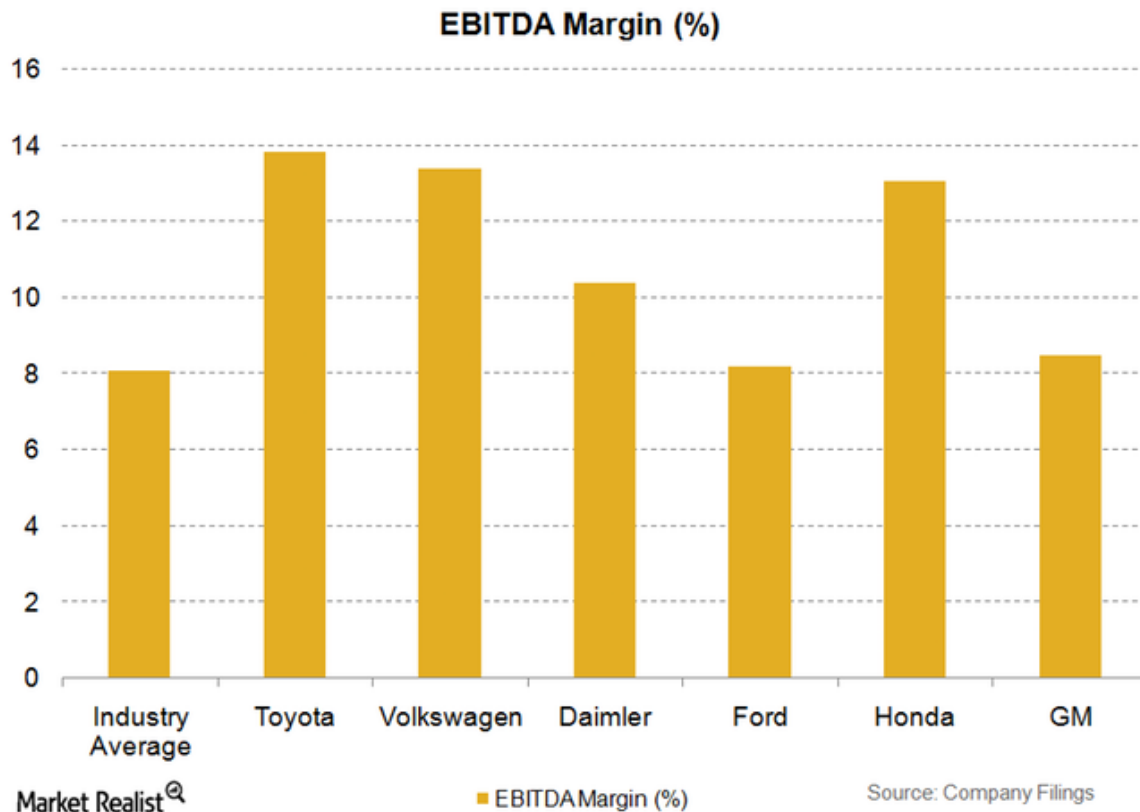


Variability among industries

Sector	EBITDA (Max)	EBITDA (Min)
Personal Goods & Retail	43.83%	4.00%
Industrial Goods & Engineering	23.27%	2.37%
Automobile and Parts	16.57%	9.10%
Chemicals	24.80%	6.37%
Gas, Water & Multiutilities	76.83%	0.00%
Transport & Travel	21.72%	4.44%
Health Care & Pharmaceuticals	42.29%	14.77%
Media, Telecommunications & Technology	38.97%	5.18%



Variability within industries



List of financial measures

	REVENUES
	OPERATING COSTS
-	(Labor, Materials, Overheads, Administration, Sales and Marketing)
=	EBITDA (Earnings Before Interests, Taxes, Depreciations, Amortizations)
-	DEPRECIATIONS
-	AMORTIZATIONS
=	EBIT (Earnings Before Interests, Taxes)



Depreciation and amortization

DEPRECIATION

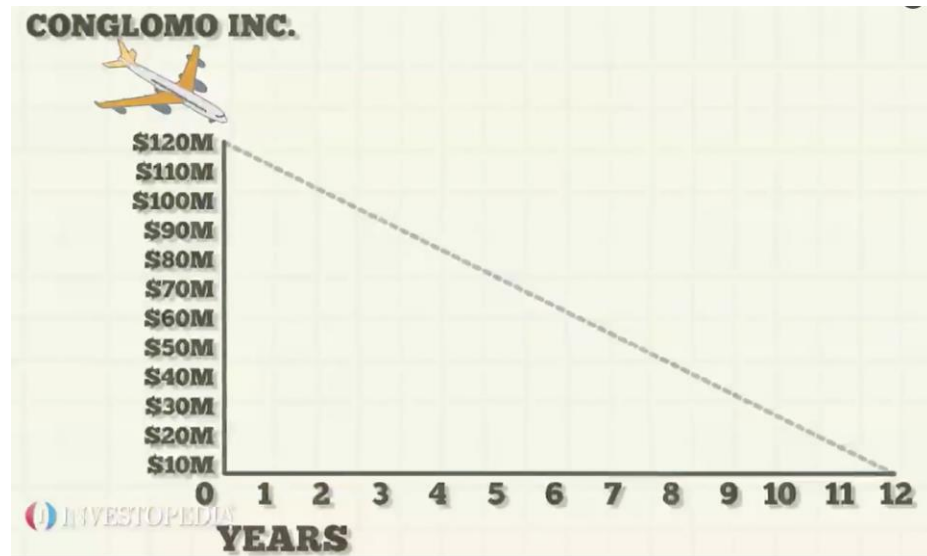
- Indicates how much of an asset's value has been used up
- Ex. A company buys a piece of equipment for \$1 million and expects it to have a useful life of 10 y. Every year, the company will expense \$100,000 of depreciation.

AMORTIZATION

- Is the spreading out of capital expenses for intangible assets over a specific period of time
- Ex. A Biotech spent \$30 million dollars on a patent with a useful life of 15 years. It would record \$2 million each year as an amortization expense.



- Conglomo acquires an airplane at \$120 mln
- Expected life of the airplane is 12 years
- Every year depreciation is \$10 mln (\$120mln/12 years)
- Conglomo does not write off the entire \$120 mln, even though the amount was paid in cash. Conglomo expenses only \$ 10mln, year after year, until year 12
- Net book value at the end of year 1 is:
 $120 - 10 = \$100$ mln
- At the end of year 12, net book value is 0 (difference between what I paid and amount of yearly depreciation)



Source: Investopedia



EBIT

- **Earnings Before Interests and Taxes**
- It is largely the operating or trading profit of the business.
- It captures the total operating income and total operating cost of the business
- It is a number of enormous significance because it is from EBIT that a company generates resources to pay interest to bankers, tax to the government, dividends to shareholders



Why the difference?

Sector	EBITDA (Max)	EBITDA (Min)	EBIT (Max)	EBIT (Min)
Personal Goods & Retail	43.83%	4.00%	40.54%	2.04%
Industrial Goods & Engineering	23.27%	2.37%	20.65%	-12.19%
Automobile and Parts	16.57%	9.10%	12.78%	4.72%
Chemicals	24.80%	6.37%	18.65%	2.26%
Gas, Water & Multiutilities	76.83%	0.00%	63.27%	0.00%
Transport & Travel	21.72%	4.44%	15.57%	-1.08%
Health Care & Pharmaceuticals	42.29%	14.77%	39.63%	11.43%
Media, Telecommunications & Technology	38.97%	5.18%	27.28%	-8.38%



List of financial measures

	REVENUES
	OPERATING COSTS
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-	DEPRECIATIONS
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=	EBIT (Earnings Before Interests, Taxes)
-	INTERESTS
=	EBT (Earnings Before Taxes)

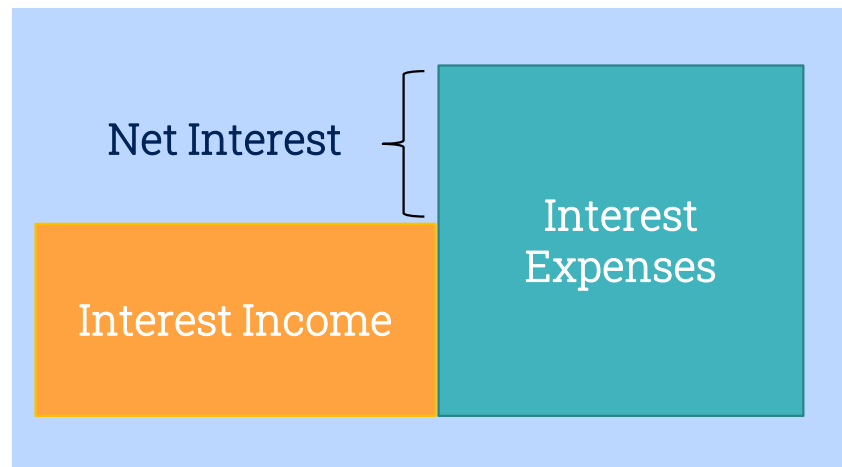


Interest expenses

— It reflects the costs of the firms to borrow from financiers (not from shareholders)

— Sometimes firms record a net figure, where:

Net interest = interest income - interests expenses



Pre-tax income

- Earnings before payment of taxes
- Numerous and diverse techniques are available to companies to avoid and/or minimize taxes that affect their reported income.



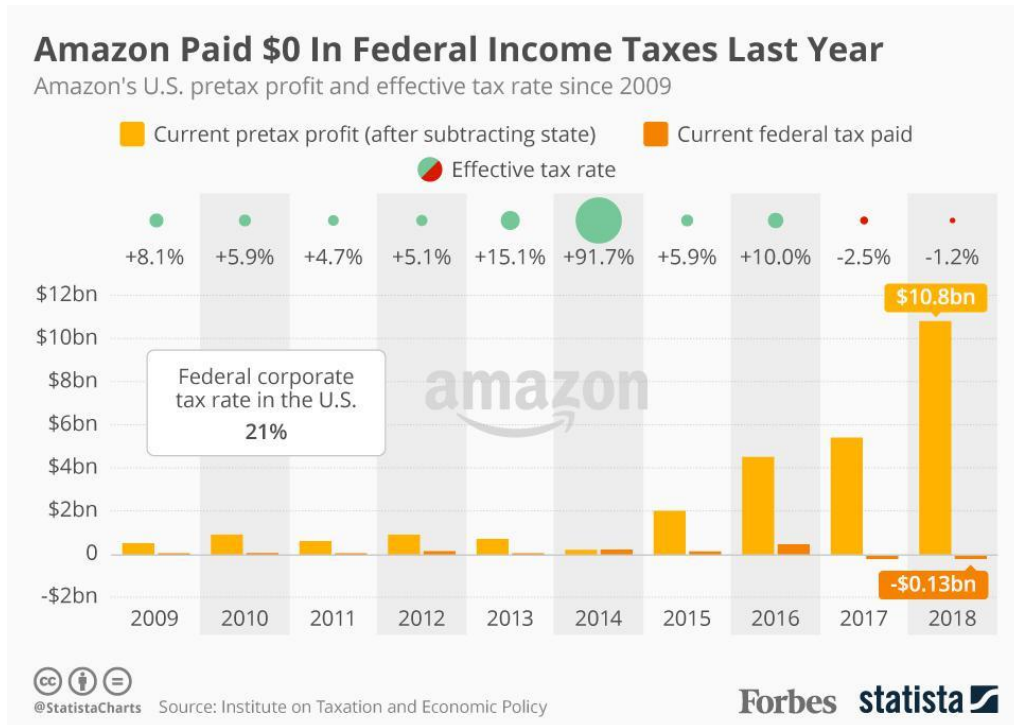
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=	NET INCOME (or Earnings After Taxes)



Income taxes

- It is an estimate, or an account that has been created to cover what a company expects to pay as taxes



Net income

- This is the bottom line, which is the most commonly used indicator of a company's profitability
- If expenses exceed income, this account caption will read as a net loss
- It gives the idea of the general wealth of the company

Company	Revenue (2019)	Net Income (2019)	Market Cap (July 2020)
Apple	\$260.2 billion	\$55.2 billion	\$1.58 trillion
Amazon	\$280.5 billion	\$11.6 billion	\$1.44 trillion
Alphabet	\$161.9 billion	\$34.3 billion	\$1.02 trillion
Microsoft	\$125.8 billion	\$39.2 billion	\$1.56 trillion
Facebook	\$70.8 billion	\$18.5 billion	\$665.04 billion
Combined	\$899.2 billion	\$158.8 billion	\$6.24 trillion

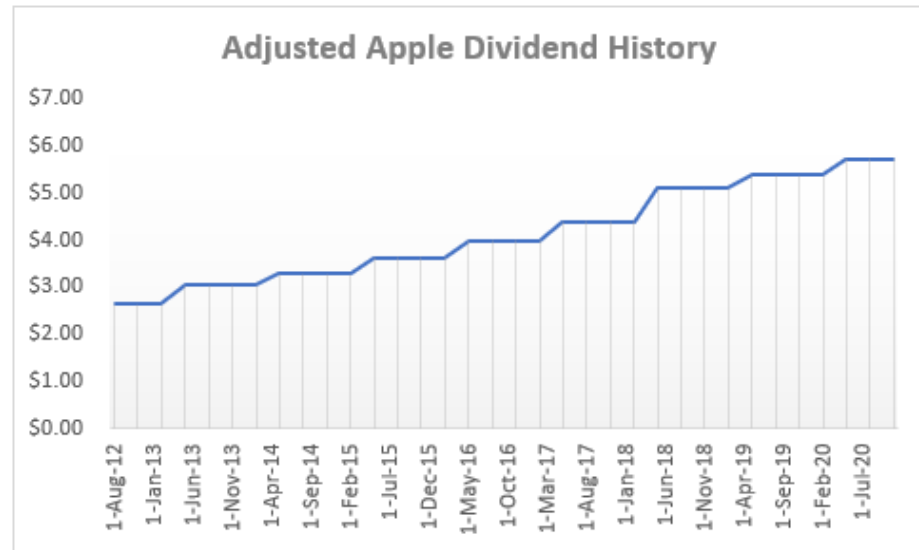
List of financial measures

	REVENUES
	OPERATING COSTS - (Labor, Materials, Overheads, Administration, Sales and Marketing)
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-	DIVIDENDS
=	RETAINED EARNINGS



Dividends

- Dividends are the part of earnings that shareholders get in cash
- Firms may decide to pay lots of dividends (e.g. AT&T) or very few dividends (e.g. Apple)



Retained earnings

- The percentage of net earnings not paid out as dividends, but retained by the company to be reinvested in its core business or to pay debt
- Often firms retain earnings to invest where they can create growth opportunities, such as buying new machinery or spending the money on more research and development
- Retained earnings are important for high-growth firms

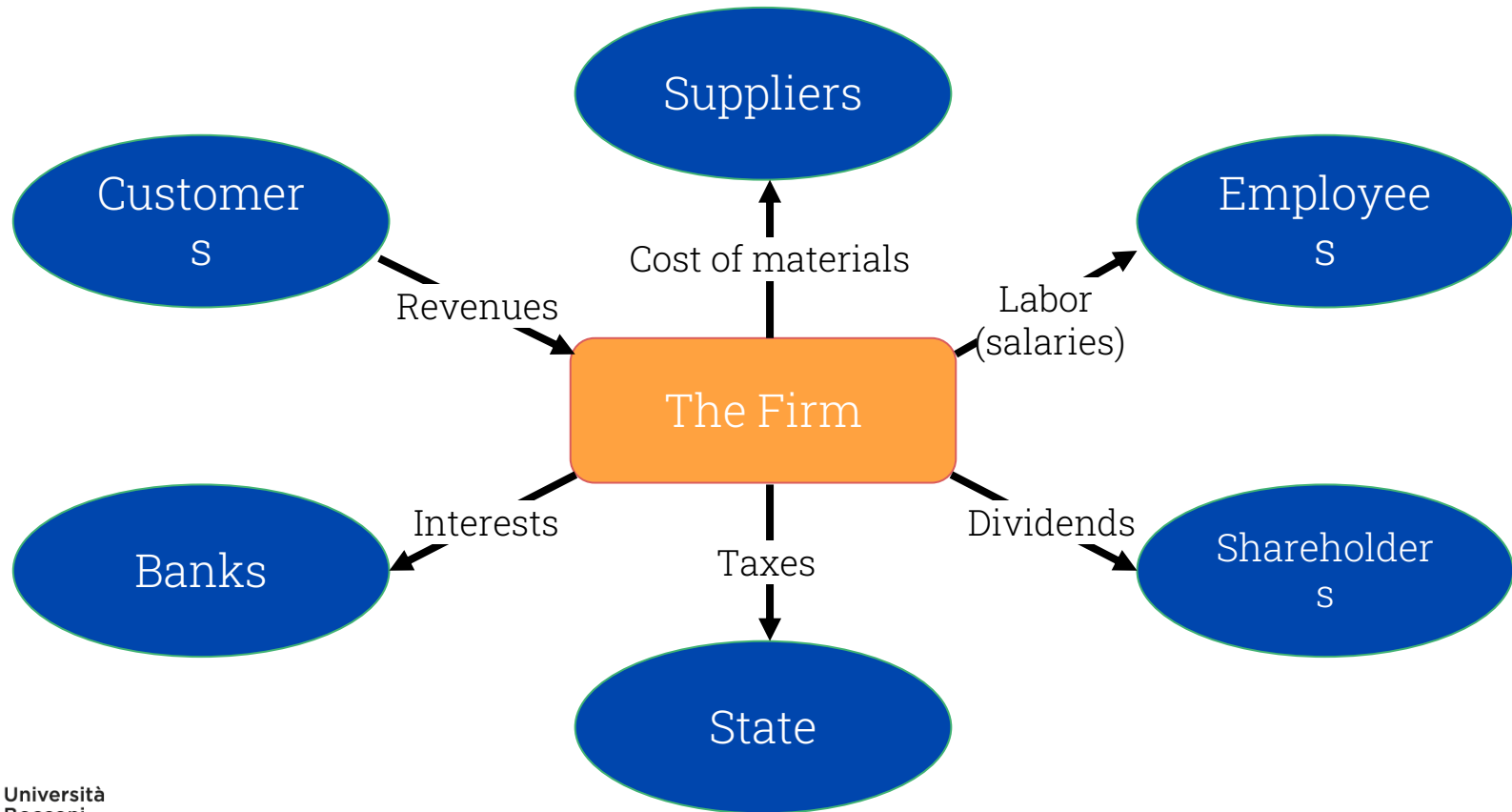


3.0

P/L AND STAKEHOLDERS



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List of financial measures

	REVENUES (CLIENTS)
-	OPERATING COSTS (Labor (EMPLOYEES), Materials (SUPPLIERS), Overheads, Administration)
=	EBITDA (Earnings Before Interests, Taxes, Depreciations, Amortizations)
-	DEPRECIATIONS
-	AMORTIZATIONS
=	EBIT (Earnings Before Interests, Taxes)
-	INTERESTS (BANKS and DEBT-HOLDERS)
=	EBT (Earnings Before Taxes)
-	TAXES (STATE)
=	NET INCOME (or Earnings After Taxes)
-	DIVIDENDS (SHAREHOLDERS)
=	RETAINED EARNINGS



4.0

RATIOS FOR STOCK MARKET INVESTORS



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Finance

Microsoft Corporation (NASDAQ:MSFT)

Add to portfolio

Company

Summary

News

Option chain

Related companies

Historical prices

Financials

57.60

+0.20 (0.35%)

Sep 30 - Close

NASDAQ real-time data - Disclaimer

Currency in USD

Range 57.34 - 57.77 Div/yield 0.39/2.71

52 week 43.92 - 58.70 EPS 2.10

Open 57.57 Shares 7.79B

Vol / Avg. 0.00/23.00M Beta 1.08

Mkt cap 450.41B Inst. own 73%

P/E 27.40

G+1

1.2k

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Recent Quotes (180 days)

chg | %

VRX 24.55 -0.500

OVS 5.15 -0.135

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Sep 26, 2016 - Sep 30, 2016 +0.17 (0.3%)



Volume (mil / 2min)



Earnings per share (EPS)

$$\frac{\text{Net income (EAT)}}{\text{No. of stocks}}$$

- It tells us the profit that shareholders get for each share held (the more the better)
- Not comparable across companies (firms may have few/many shares of low/large value)
- More informative to look at changes over time: firm profitability growth



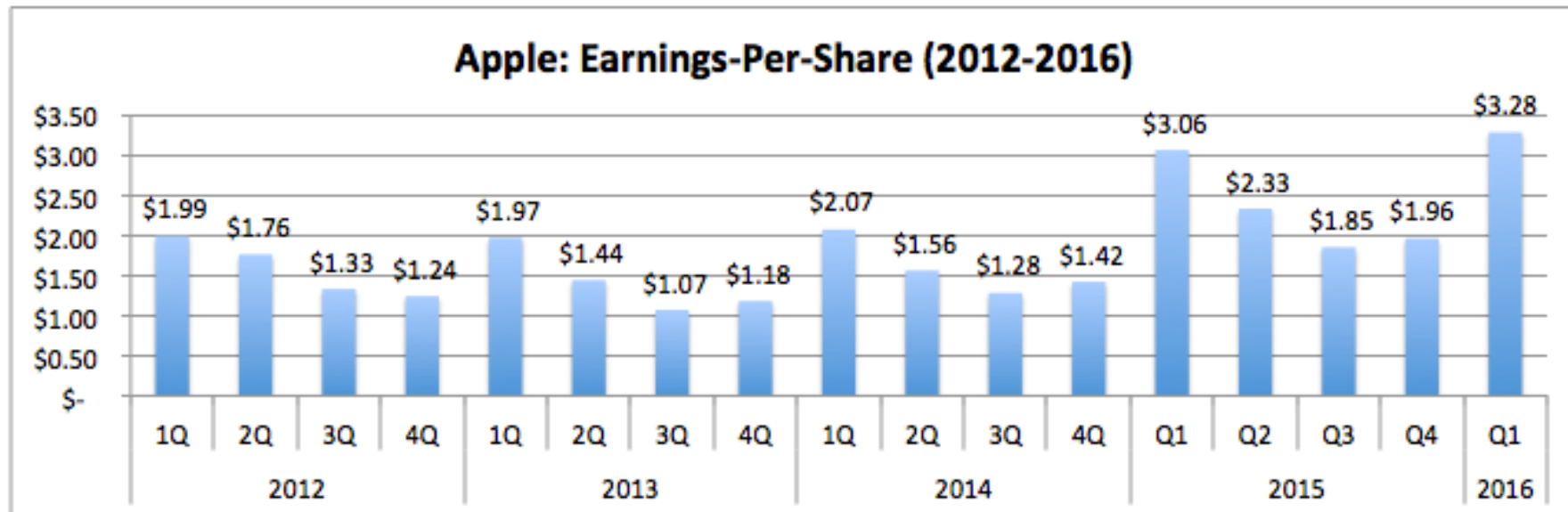
EPS of some companies (September 30, 2016)

—As you see, EPS is **not comparable across companies** because it depends on how many shares a company issued

	Net income	Common shares	EPS
Amazon	\$ 1.901 B	474.04 M	\$ 4.01
Apple	\$ 46.138 B	5.390 B	\$ 8.56
Facebook	\$ 4.849 B	2.320 B	\$ 2.09
Google	\$ 7.754 B	294.84 M	\$ 26.30



But growth in EPS in a company is an important measure of progress



Price to earnings ratio (P/E)

$$\frac{\text{Share price}}{\text{Earning per share}}$$

- Reflects the number of years of earnings to pay back the purchase price.
- High P/E means that investors are paying more for each share of current net income → low likelihood of hostile takeovers, financial means (shares) to make acquisition.
- High P/E means that the market loves the company.



P/E of some companies (September 30, 2016)

- P/E is **comparable across companies** and is an indicator of the opinion of investors about the potential growth in the profits of a company (high P/E implies high expected growth).

	Share price	EPS	P/E
Amazon	\$ 837.31	\$ 4.01	208.79
Apple	\$ 113.05	\$ 8.56	13.20
Facebook	\$ 128.27	\$ 2.09	61.44
Google	\$ 804.06	\$ 26.30	30.57



Market to book ratio (MtB)

$$\frac{\text{Market cap}}{\text{Owners' funds}}$$

- What does it reflect?
- What is a firm with $\text{MtB} > 1$?
 - And $\text{MtB} < 1$?



Market to book ratios of some companies (September 30, 2016)

	Market Cap	Book value of equity	Market to book
Amazon	\$ 400.900 B	\$ 16.538 B	24.24
Apple	\$ 613.890 B	\$ 126.541 B	4.85
Facebook	\$ 368.87 B	\$ 50.383 B	7.32
Google	\$ 544.37 B	\$ 127.879 B	4.26



Dividend per share

$$\frac{\text{Dividends paid}}{\text{No. of stocks}}$$

- Like EPS, but with Dividends in place of Net Income.
- DPS determines the dividend earned by each shareholder. For example, the last DPS of Microsoft was \$ 0.39. This means that if Bill has 1 M shares, he gets $(\$ 0.39 \times 1 \text{ M}) = \$ 390,000$.
- Increasing DPS is a great way for a company to signal strong performance to its shareholders
Many firms like to keep DPS constant (or make it regularly grow) over the years. Doing so demonstrates management's trust in the ability of the firm to generate steady or growing profits.



Payout ratio

$$\frac{\text{Dividends paid}}{\text{Net income (EAT)}}$$

OR

$$\frac{\text{DPS}}{\text{EPS}}$$

- The fraction of earnings that is paid out to shareholders.
- Microsoft: the company pays four dividends a year → $\text{DPS} = \$ 1.56$ and $\text{EPS} = \$ 2.10$. So the pay-out ratio is $\$ 1.56 / \$ 2.10 = 74.3\%$. The remaining 25.7% is retained (retained earnings) and added to owners' funds.
- Some companies leave DPS free to vary and try to keep the pay-out ratio constant (e.g. they pay 40% of the earnings out to shareholders, no matter the absolute amount).



THANKS.



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